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A

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D

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G

H

NATIVE SOURCE. Power development; safety/PHY/PCB/bench qualification remain OPEN.

01_carrier_interface

10_input_protection

20_positive_rails

30_negative_reference

40_power_supervision

03_peripheral_boundaries

04_dac_0_3

05_dac_4_7

06_analog_outputs

41_power_status

File: 01_carrier_interface.kicad_sch

File: 10_input_protection.kicad_sch

File: 20_positive_rails.kicad_sch

File: 30_negative_reference.kicad_sch

File: 40_power_supervision.kicad_sch

File: 03_peripheral_boundaries.kicad_sch

File: 04_dac_0_3.kicad_sch

File: 05_dac_4_7.kicad_sch

File: 06_analog_outputs.kicad_sch

File: 41_power_status.kicad_sch

Sheet: /

File: servohl_io_revB.kicad_sch

Title: ServoHIL Rev.B – native editable single–SoC expansion

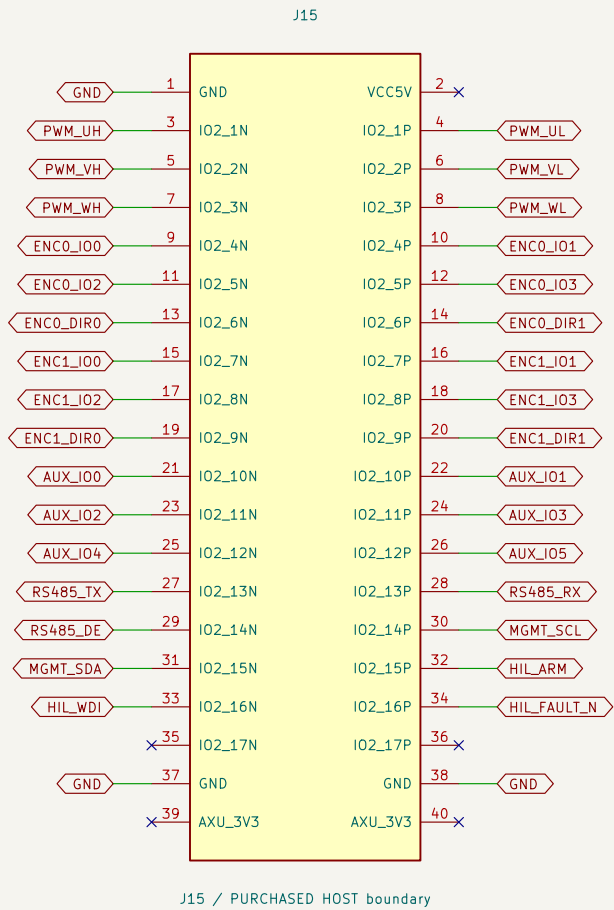
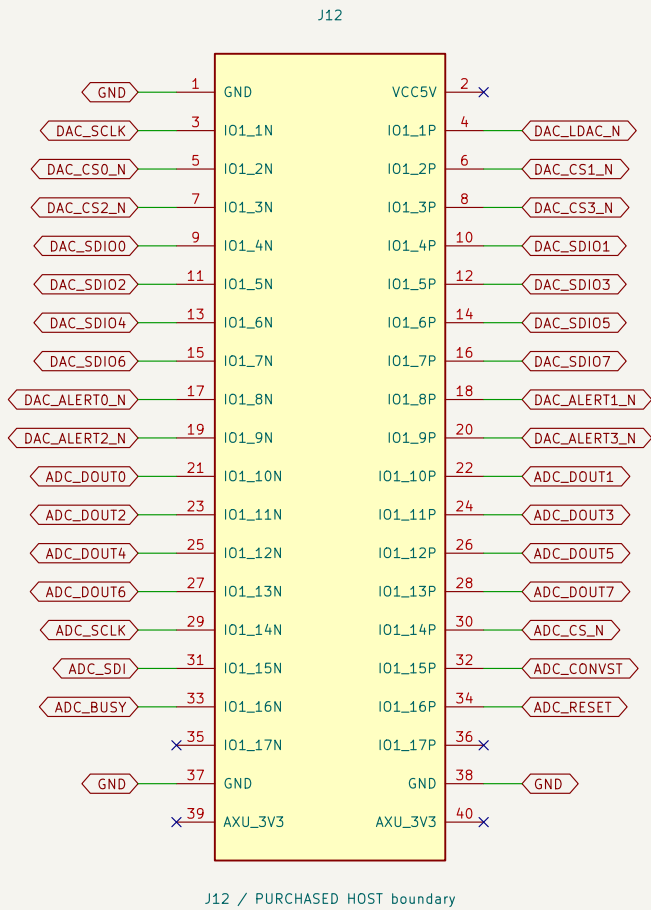
Size: A2

Date:

Rev: B–power–development

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Connector power contacts remain NC. 64 assigned GPIO; unused GPIO also NC. No secondary compute chip.

J12 and J15 are ORIGINAL AXU2CGB interfaces, not AXU2CGB–I or –E. No board–power backfeed through contacts.

Signal–pin backfeed, output enable and bank power sequencing STILL REQUIRE electrical review.

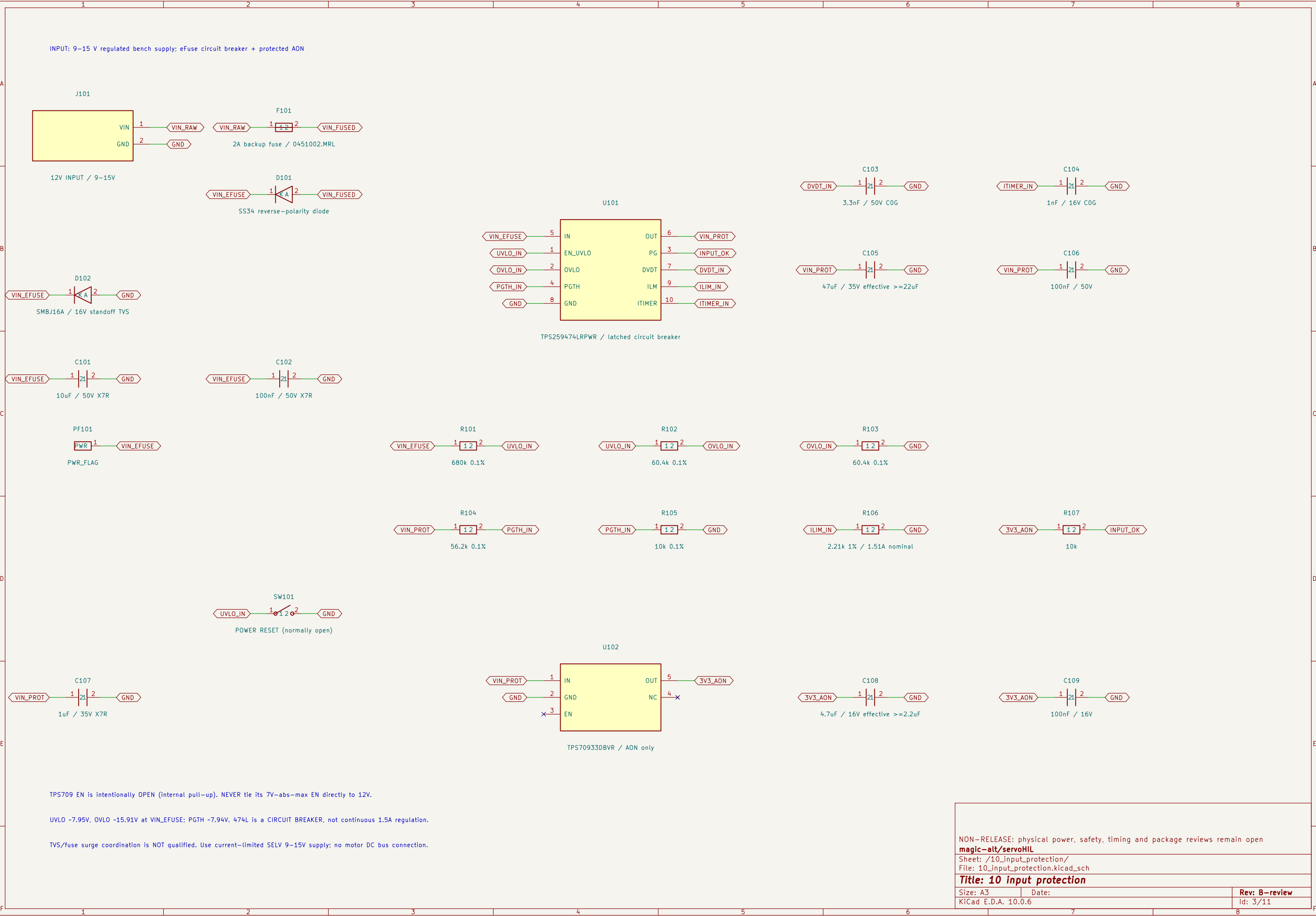
NON–RELEASE: physical power, safety, timing and package reviews remain open

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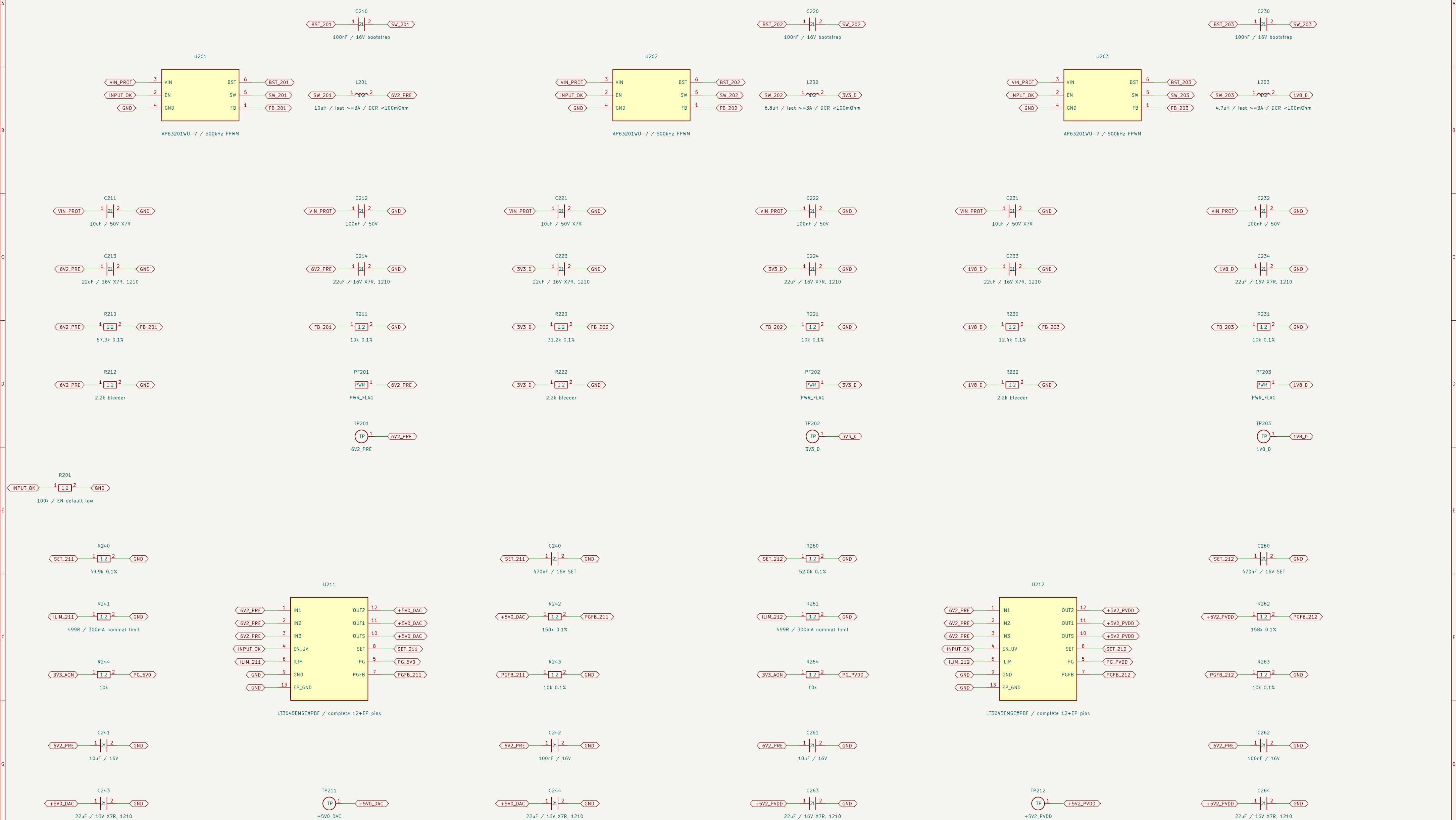
Sheet: /01_carrier_interface/
File: 01_carrier_interface.kicad_sch

Title: 01 carrier interface

Size: A3	Date:	Rev: B–review
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POSITIVE POWER: 6.184V preregulator / 3.296V / 1.792V + LT3045 DAC rails



Bucks: AP63201 internal compensation; no invented COMP network. Cout requires >=20uF EFFECTIVE per buck after DC--bias/temperature.

Power budgets: 6V2_PRE 0.50A, 3V3_D 0.30A, 1V8_D 0.30A; +5V0_DAC 0.20A, +5V2_PVDD 0.12A. NOT a 2A--per--rail board rating.

LT3045 OUTS and SET require Kelvin routing. MSE EP13=GND, +5V0 PG trip -4.80V; +5V2 PG trip -5.04V. Layout/thermal/loop response OPEN.

NON--RELEASE: physical power, safety, timing and package reviews remain open

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Sheet: /20_positive_rails/

File: 20_positive_rails.kicad_sch

Title: 20 positive rails

Size: A2

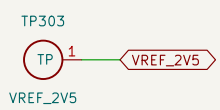
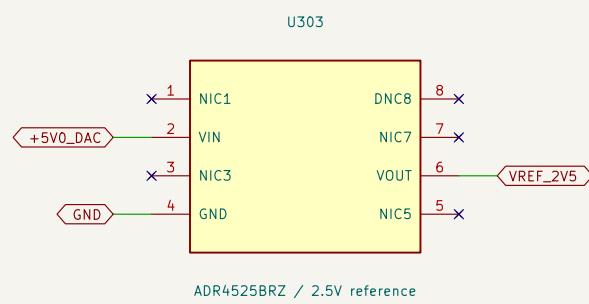
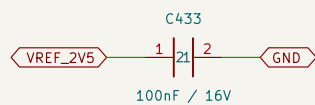
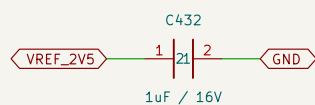
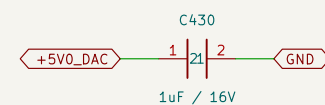
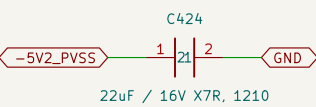
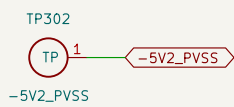
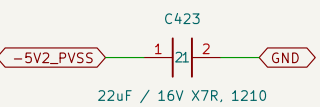
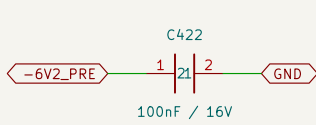
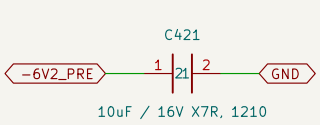
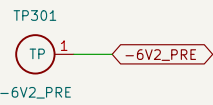
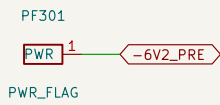
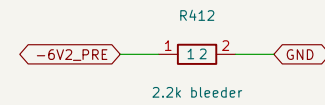
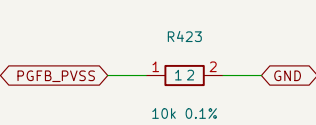
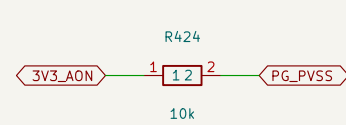
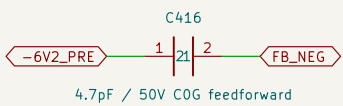
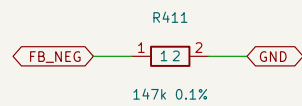
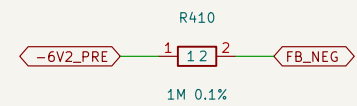
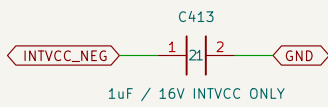
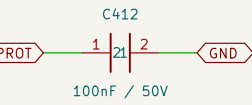
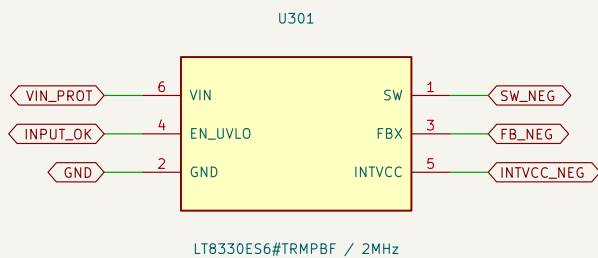
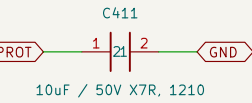
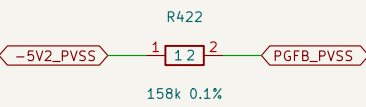
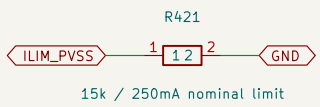
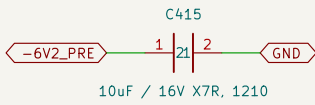
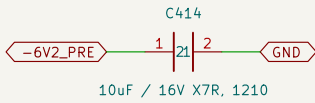
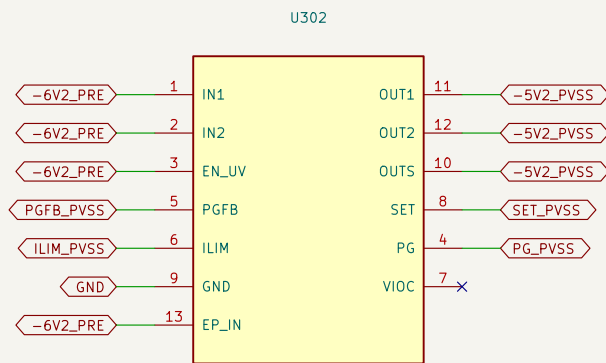
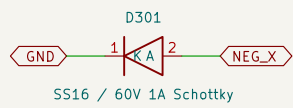
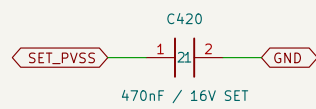
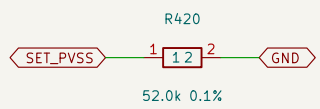
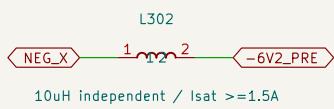
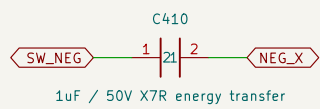
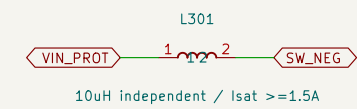
Date:

Rev: B--review

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Id: 4/11

NEGATIVE POWER / VREF: LT8330 Cuk -> LT3094 -> DAC; ADR4525 external reference



Inverting stage topology: VIN -> L301 -> SW -> C410 -> NEG_X -> L302 -> negative OUT; D301 ANODE=NEG_X, CATHODE=GND.

Independent 10uH inductors, not shorted/parallel windings. C410 sees VIN + |VOUT|. Target -6.24V; negative continuous budget 0.12A.

LT3094 MSE EP13 and EN3 go to negative IN, NOT GND. SET sinks 100uA. PG trip --5.04V. Kelvin OUTS/SET and >=10uF effective Cout required.

ADR4525 pin8 DNC; configure DAC external--reference mode before use. Negative--loop ripple/thermal/startup and reference fanout remain bench gates.

NON-RELEASE: physical power, safety, timing and package reviews remain open

~~magic~~-alt/servoHIL

Sheet: /30_negative_reference/

File: 30_negative_reference.kicad_sch

Title: 30 negative reference

Size: A2

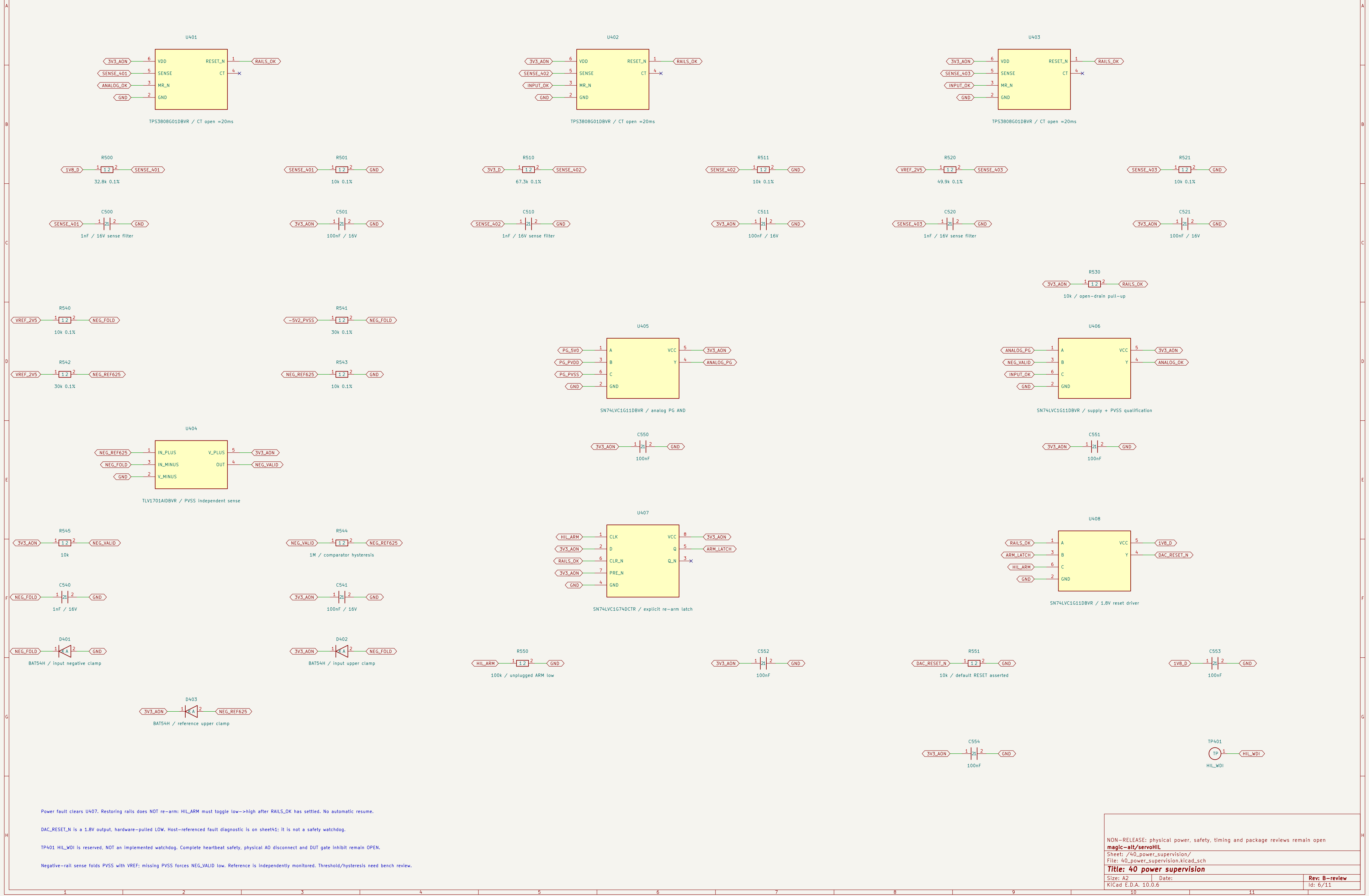
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 5/11

POWER QUALIFICATION: monitored rails, negative-rail loss detection and latched DAC reset release



Power fault clears U407. Restoring rails does NOT re-arm: HIL_ARM must toggle low-->high after RAILS_OK has settled. No automatic resume.

DAC_RESET_N is a 1.8V output, hardware-pulled LOW. Host-referenced fault diagnostic is on sheet41; it is not a safety watchdog.

TP401 HIL_WDI is reserved, NOT an implemented watchdog. Complete heartbeat safety, physical AO disconnect and DUT gate Inhibit remain OPEN.

Negative-rail sense folds PVSS with VREF; missing PVSS forces NEG_VALID low. Reference is independently monitored. Threshold/hysteresis need bench review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-alt/servoHIL

Sheet: /40_power_supervision/

File: 40_power_supervision.kicad_sch

Title: 40 power supervision

Size: A2

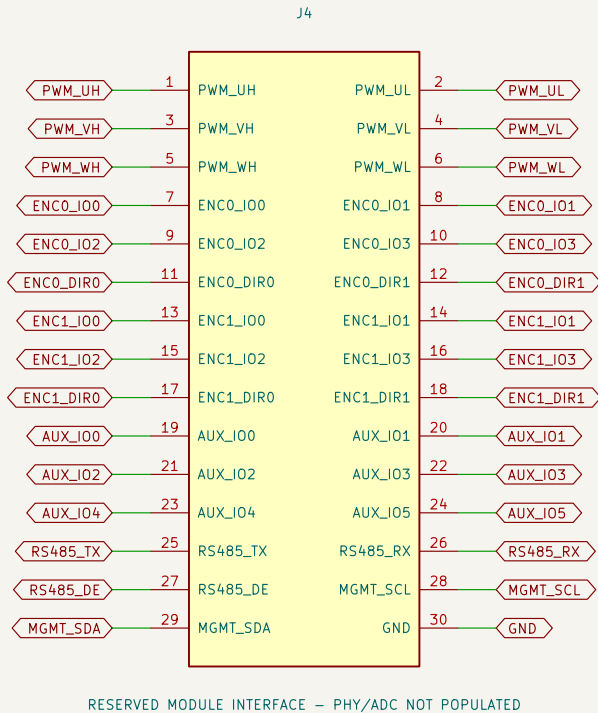
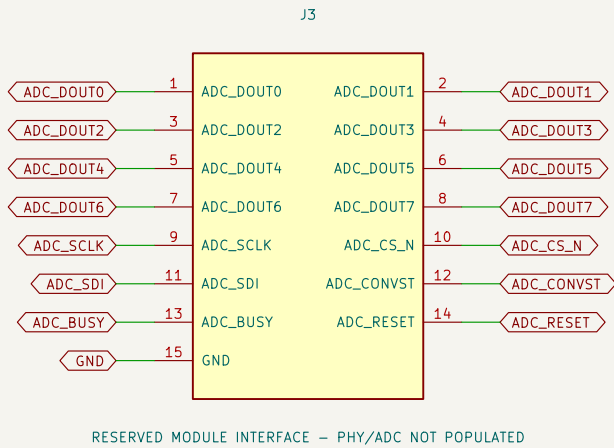
Date:

Rev: B-review

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Id: 6/11

03 – ADC and digital PHY integration interfaces (reserved)



Eight AI channels are an interface reservation. No 1 MSPS verification is implied; 1.8V serial timing must be closed.

Each encoder uses one selected profile; ABZ/SSI/BISS/SPI are NOT simultaneously independent ports.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-alt/servoHIL

Sheet: /03_peripheral_boundaries/
File: 03_peripheral_boundaries.kicad_sch

Title: 03 peripheral boundaries

Size: A3	Date:	Rev: B-review
KiCad E.D.A. 10.0.6		Id: 7/11

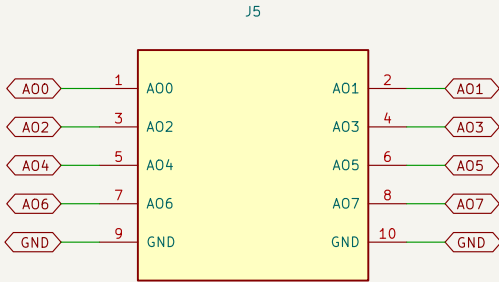


3



Id: 8/11

06 – Eight analog outputs to reviewed DUT adapter



A00..A07 – NOT MCU–safe without adapter

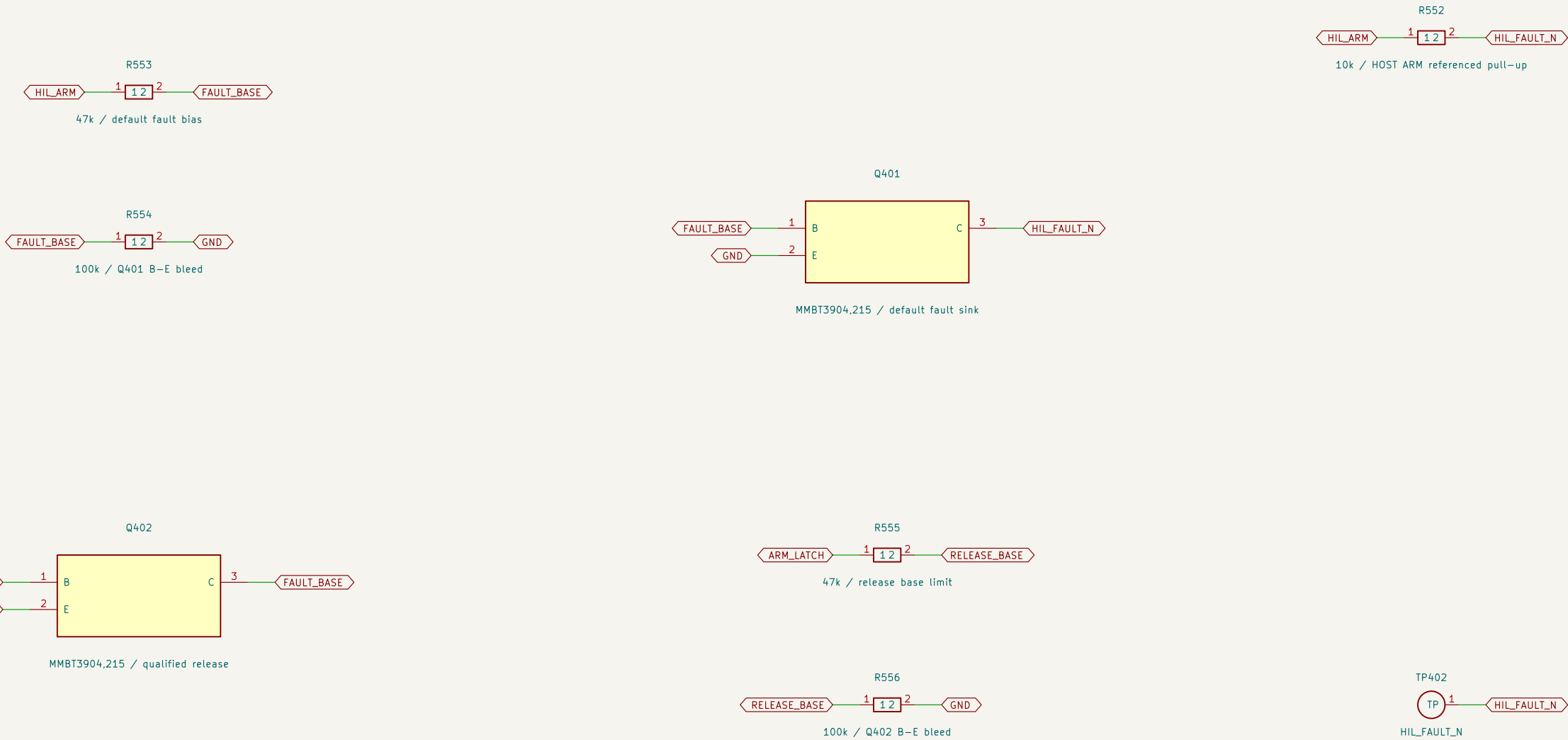
Default metadata: Ia, Ib, Ic, Vbus, torque, temperature, AUX0, AUX1. Electrical channels are equivalent.

DUT adapter supplies scaling/clamping and hardware inhibition. No direct +/-5V connection to MCU ADC.

NON–RELEASE: physical power, safety, timing and package reviews remain open
~~magic–alt/servoHIL~~
Sheet: /06_analog_outputs/
File: 06_analog_outputs.kicad_sch

Title: 06 analog outputs		
Size: A3	Date:	Rev: B–review
KiCad E.D.A. 10.0.6		Id: 10/11

HOST-REFERENCED POWER STATUS: no expansion-rail pull-up on HIL_FAULT_N



HIL_ARM is generated by the powered HOST, NOT an expansion rail. Q401 defaults ON while HOST requests ARM.

Only a power-qualified ARM_LATCH turns Q402 on and releases Q401. Board power lost -> Q402 off -> Q401 reports fault.

HIL_FAULT_N=HIGH means power-qualified AND armed; LOW also means disarmed. At HOST off, no expansion supply drives this pin high.

Nominal fault sink <0.33mA and HOST ARM total load <0.4mA. Check transient/partial-power behavior physically; not certified safety.

Other direct DAC/ADC pins STILL require power-off backfeed protection. This diagnostic alone does not authorize hot plugging.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-alt/servoHIL

Sheet: /41_power_status/

File: 41_power_status.kicad_sch

Title: 41 power status

Size: A3

Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 11/11